

SUPPLEMENTARY INFORMATION

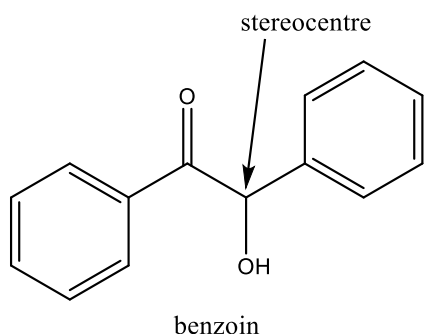
Section A

Experiment A1:

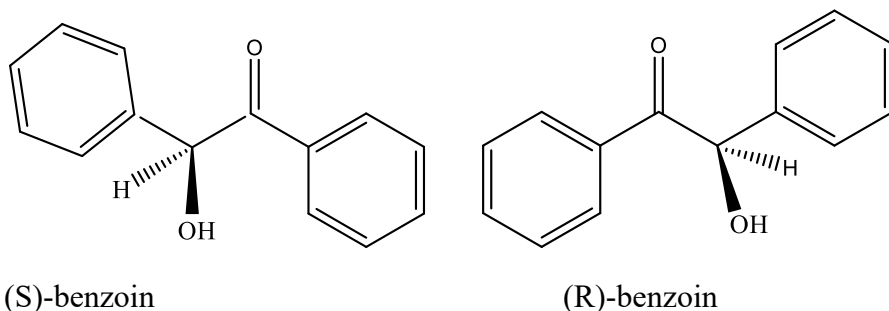
Question: The benzoin produced in this experiment contains a stereocentre, but the product itself is not optically active. Explain.

Answer:

Benzoin does contain a single stereocentre: the carbon bearing the hydroxyl group.

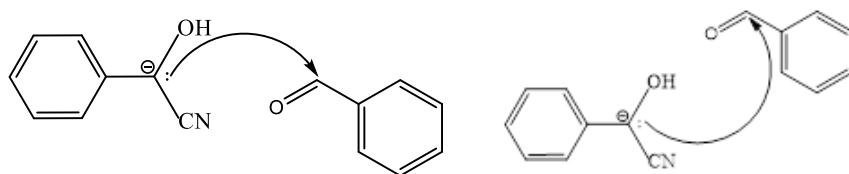


Benzoin can in principle exist as a pair of enantiomers, (R)- and (S)-benzoin.



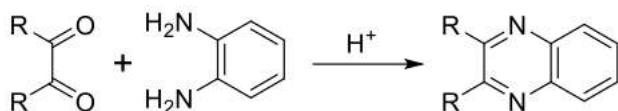
The reason the product shows no optical activity is that it is obtained as a racemate — a 50:50 mixture of the two enantiomers — and the equal and opposite rotations of the two enantiomers exactly cancel, giving a net optical rotation of zero (optical inactivity by external compensation).

That it must be racemic follows from the reaction itself. Both the starting material (benzaldehyde) and the catalyst (cyanide ion) are achiral, so there is no chiral influence anywhere in the system. The stereocentre is created in the step where the resonance-stabilised carbanion (intermediate 1) adds to the carbonyl carbon of the second molecule of benzaldehyde. That carbonyl carbon is planar (sp^2), so the nucleophile can attack with equal probability from either face. Attack on one face generates the (R) product and attack on the other gives the (S) product; because nothing energetically distinguishes the two faces, the enantiomers are produced in exactly equal amounts.



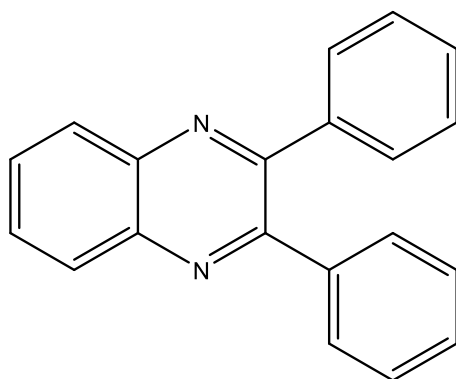
Experiment A2:

Question: 1,2-Dicarbonyl compounds, such as benzil, can be characterized by reaction with *o*-phenylenediamine to form a substituted quinoxaline.



a. Write the structure for the quinoxaline derivative obtained when benzil is the reactant. Do you think this compound would be coloured? If so, why?

Answer:



2,3-diphenylquinoxaline

Yes, it's coloured — pale yellow.

Reason: Extended π conjugation. The quinoxaline core is a fully aromatic 10- π -electron bicycle, and the two phenyls extend that conjugation further. The more extended the conjugated π system, the smaller the HOMO–LUMO gap. When that gap drops to the energy of visible light, the molecule absorbs in the blue–violet region and we see the transmitted/reflected yellow. (The two phenyls can't sit perfectly coplanar with the quinoxaline — they twist out of plane to relieve mutual steric clash — which limits the extension and keeps the yellow pale rather than deep.)

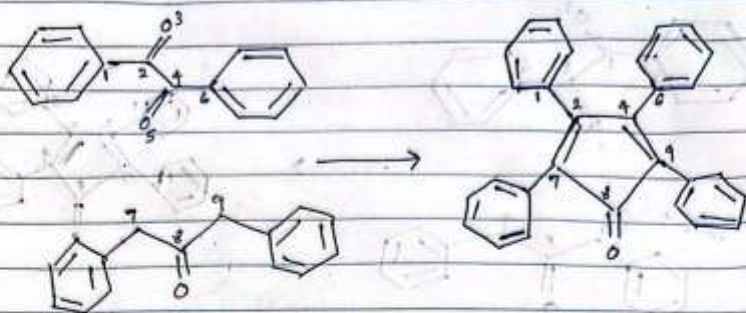
b. Suggest a plausible mechanism for the formation of the quinoxaline compounds based on the reaction scheme shown above.

Experiment A3:

Question: 3. Outline a complete mechanistic sequence to account for the formation of the tetraphenylcyclopentadienone compound.

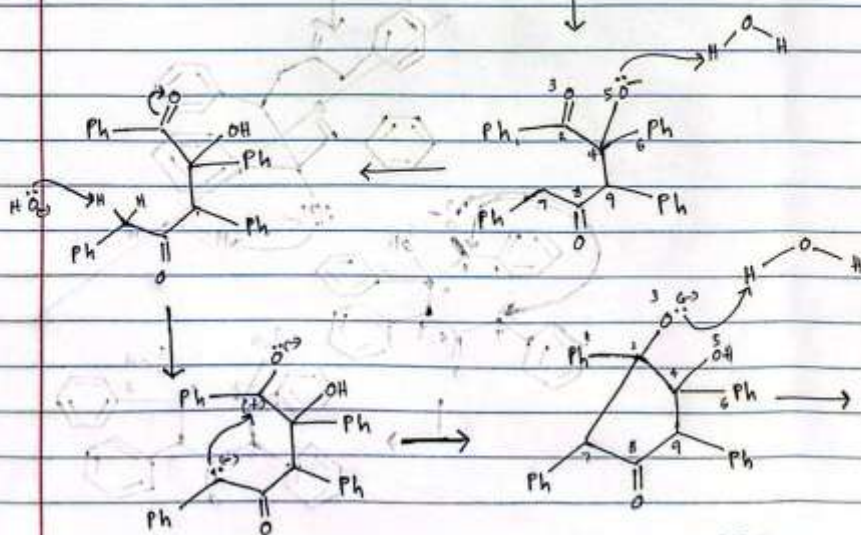
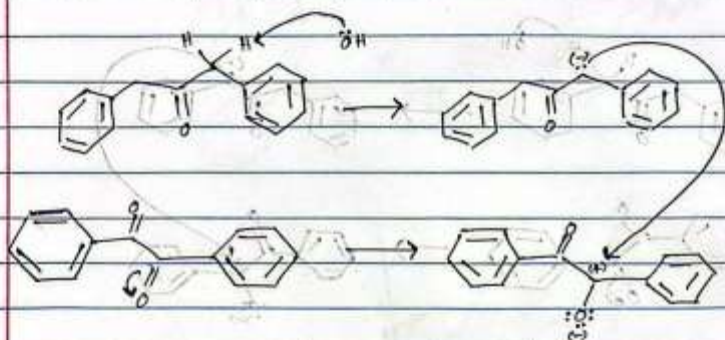
Answer:

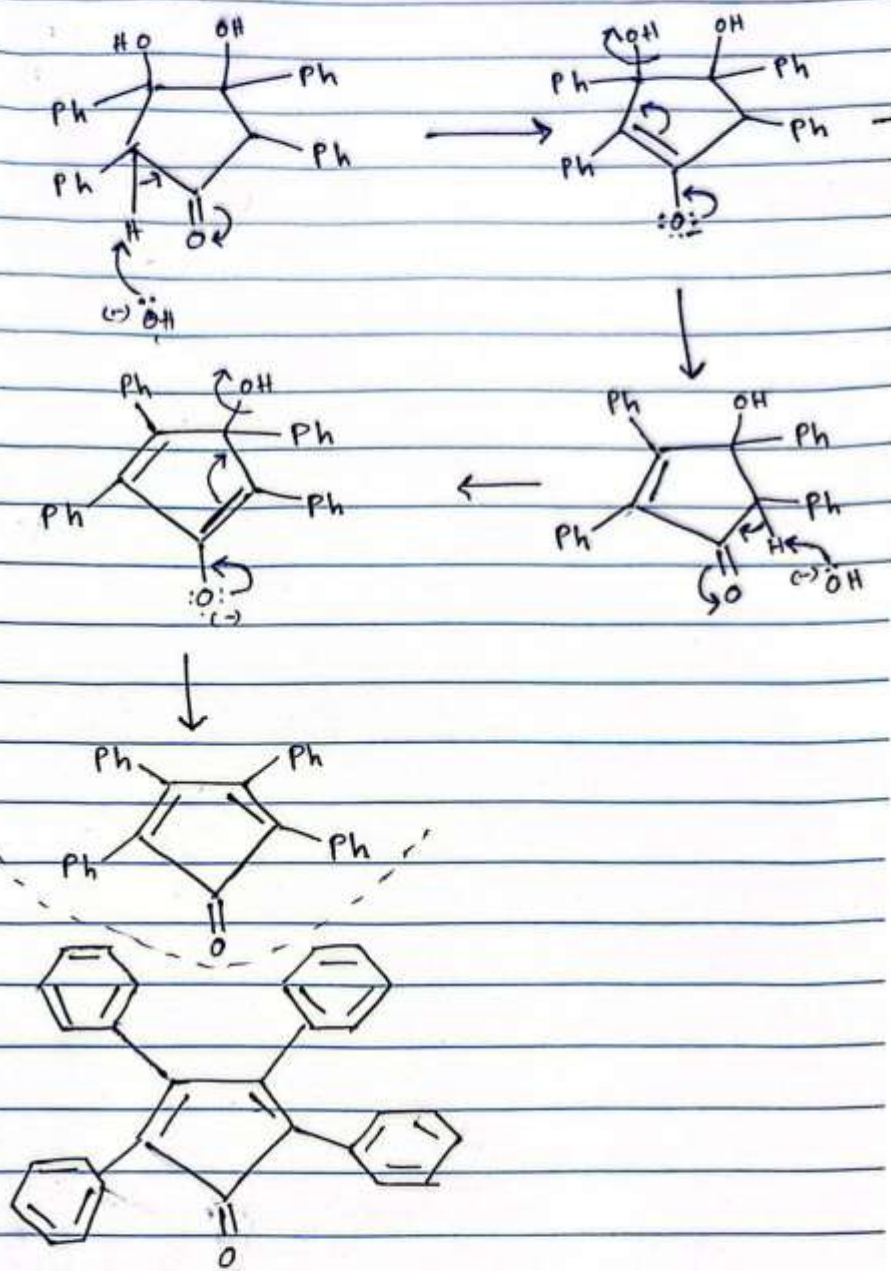
Answer : Q3



Bonds broken $\rightarrow$ 2-3, 4-5

Bonds made $\rightarrow$ 2-7, 4-9





Experiment B1:

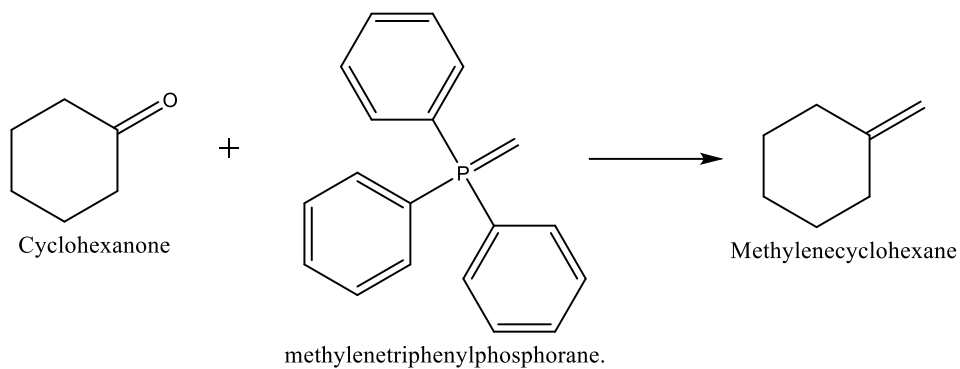
Question: Give the structure of the phosphorus ylide and carbonyl compound you might use to prepare the following alkenes:

a. Methylenecyclohexane

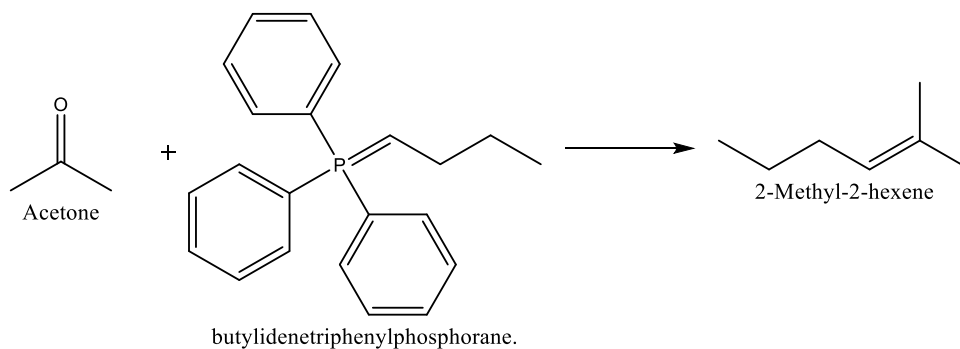
b. 2-Methyl-2-hexene

c. $\text{C}_6\text{H}_5\text{CH}=\text{C}(\text{CH}_3)_2$

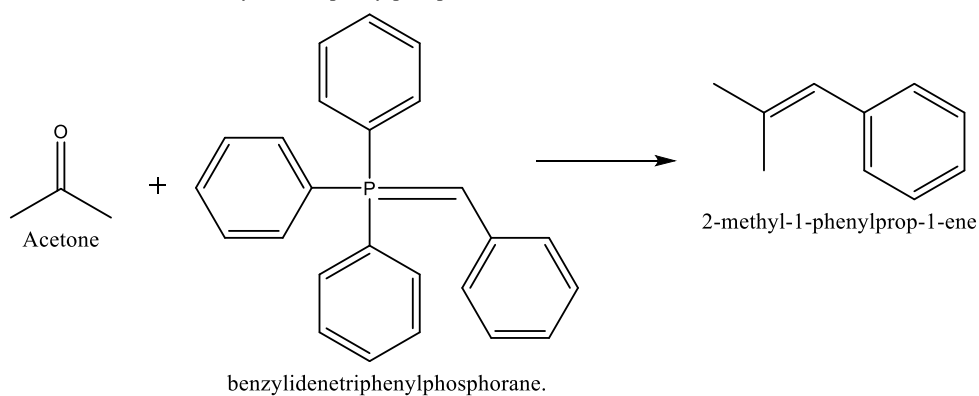
Answer:



a.



b.

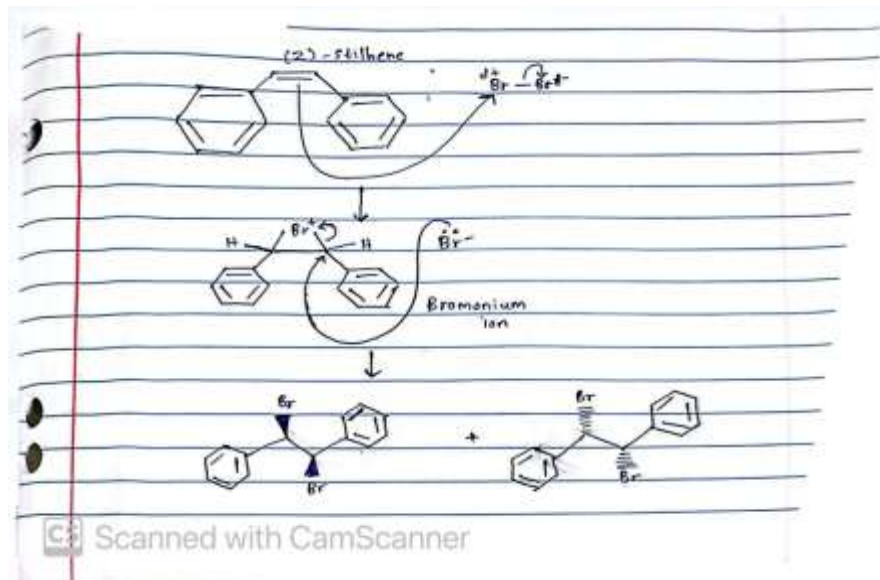


c.

Experiment B2:

Question: a. Using suitable structures, draw the reaction mechanism and products for the addition reaction of bromine to (Z)-stilbene.

Answer:

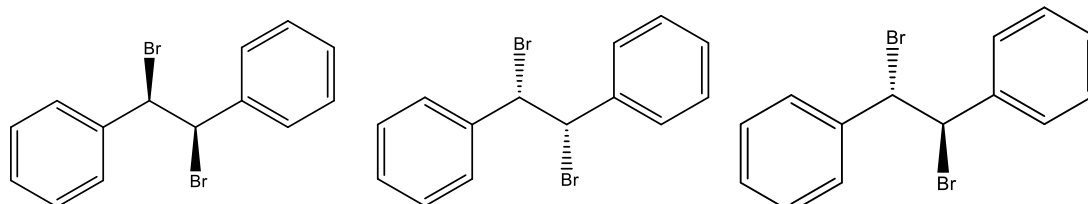


b. Are the products different from those made in this experiment? If so, how? What is the stereochemical relationship between the dibromide products formed from the two reactions (Z versus E)?

Answer:

Yes. The products are different from the meso product made in the experiment. The *E*-reaction gave one achiral compound (meso); the *Z*-reaction gives a racemic mixture of two chiral enantiomers. The two products are diastereomers of each other.

Anti addition means the two Br's end up on opposite faces. In *E*-stilbene, the two Ph's are also on opposite sides, so the result is symmetric (one Br on each side, one Ph on each side → meso). In *Z*-stilbene, the two Ph's are on the same side, but the Br's still end up on opposite sides — that combination breaks the internal symmetry and gives a chiral product. Because the starting alkene is itself achiral (planar) and bromonium formation has no chiral preference, we get equal *R,R* and *S,S* → racemic.



(1*R*,2*R*)-1,2-dibromo-1,2-diphenylethane (1*S*,2*S*)-1,2-dibromo-1,2-diphenylethane meso-1,2-dibromo-1,2-diphenylethane

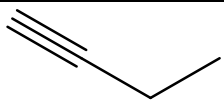
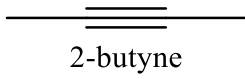
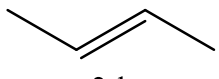
Experiment B3:

Question: Use IR tables to locate the absorption bands of the stretching frequencies of the alkyne C-H bond, the alkyne C≡C bond, and the alkene C-H bond. Using this IR data, explain how you would distinguish between 1-butyne, 2-butyne, and 2-butene.

The three relevant absorption bands:

Bond	Typical wavenumber
≡C-H stretch (terminal alkyne)	3260–3340 cm ⁻¹
C≡C stretch	2100–2260 cm ⁻¹
=C-H stretch (alkene)	3010–3100 cm ⁻¹

The three molecules and what their spectra look like:

Compound	Structure	≡C-H (~3300)	=C-H (~3030)	C≡C (~2150)
1-butyne	 1-butyne	Present, sharp	Absent	Present, medium
2-butyne	 2-butyne	Absent	Absent	Absent (IR-silent)
2-butene	 trans-2-butene	Absent	Present	Absent

2-Butyne is a symmetric internal alkyne: the C≡C sits at the centre of symmetry of the molecule, with identical methyl groups on each side. The C≡C stretching motion is a symmetric vibration that does not change the dipole moment of the molecule (it stays zero throughout the stretch). IR absorption requires a change in dipole moment, so this vibration is IR-inactive — the C≡C bond is essentially invisible in the IR spectrum of 2-butyne. (It's Raman-active, but that's a different technique.)

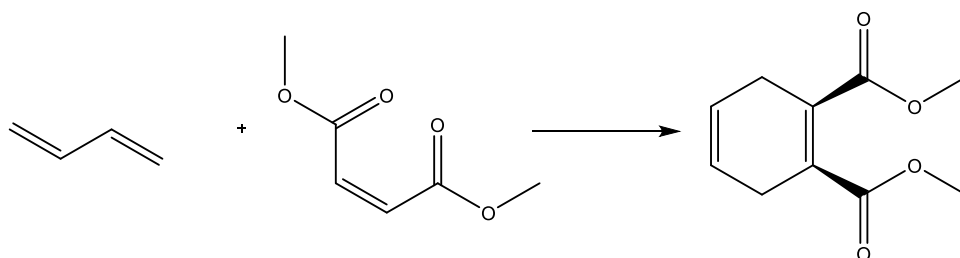
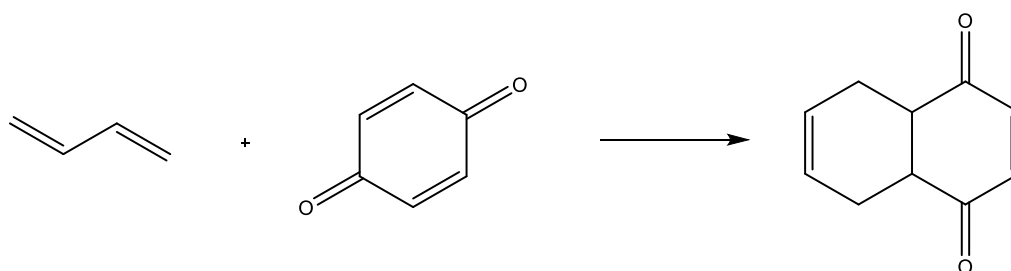
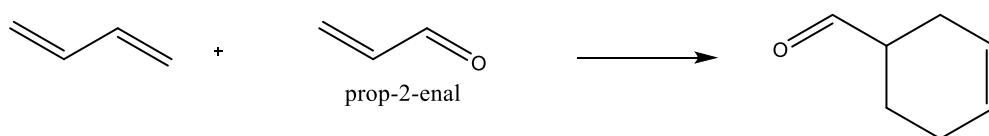
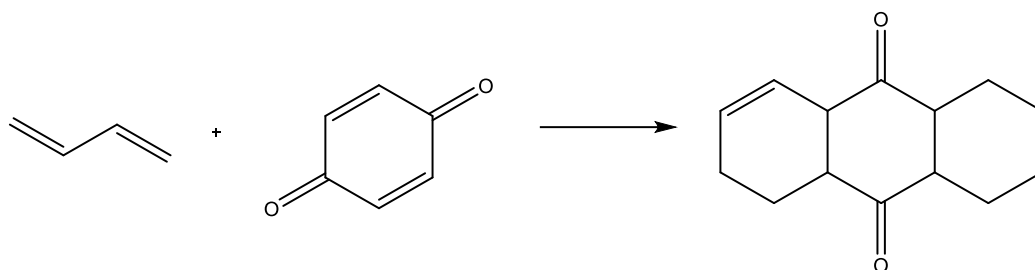
1-Butyne is a terminal alkyne. Its spectrum shows the sharp ≡C-H stretch around 3300 cm⁻¹ and the C≡C stretch around 2120 cm⁻¹. There is no absorption above 3000 from =C-H.

2-Butene is an alkene. There is no ≡C-H stretch at 3300 cm⁻¹ and no C≡C stretch in the 2100–2260 region. The diagnostic feature is the alkene =C-H stretch just above 3000 cm⁻¹ (~3020).

Experiment AB4:

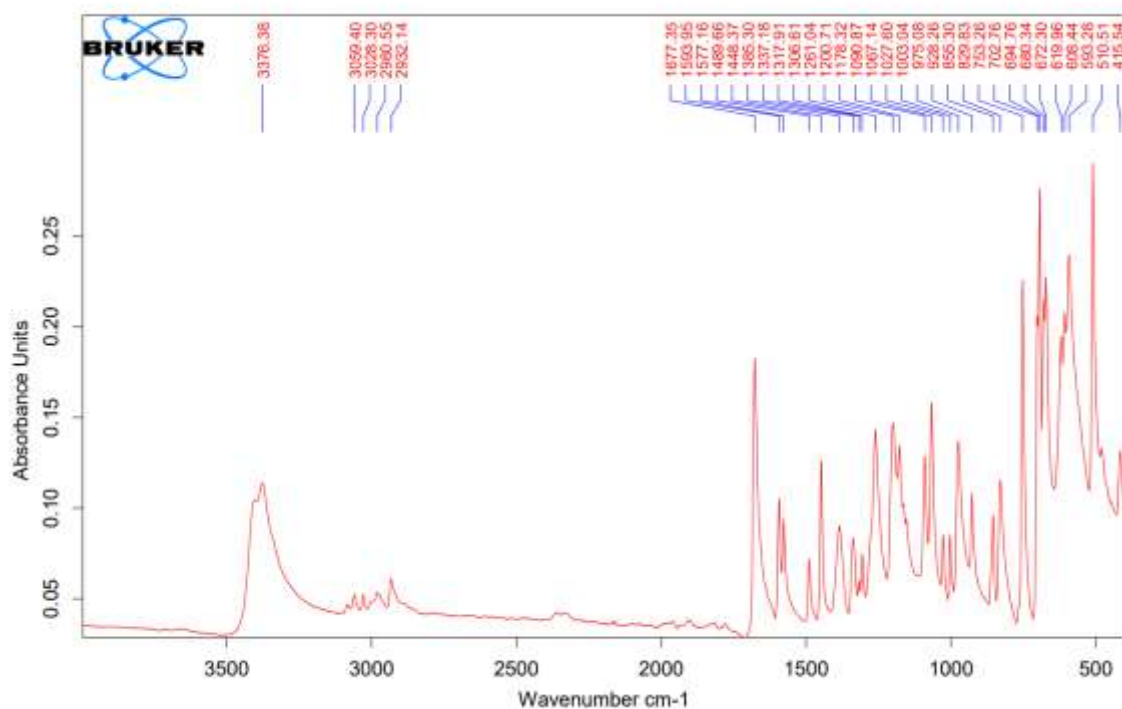
Question: 7. What starting materials would you use to prepare each of the following compounds by the Diels-Alder reaction?

Answer:



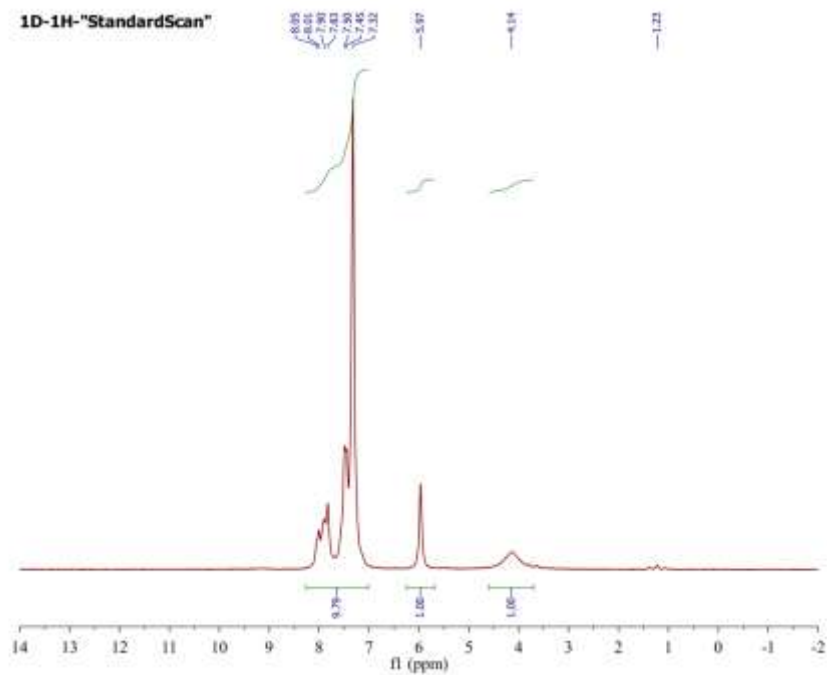
Section B

Experiment A1: Benzoin



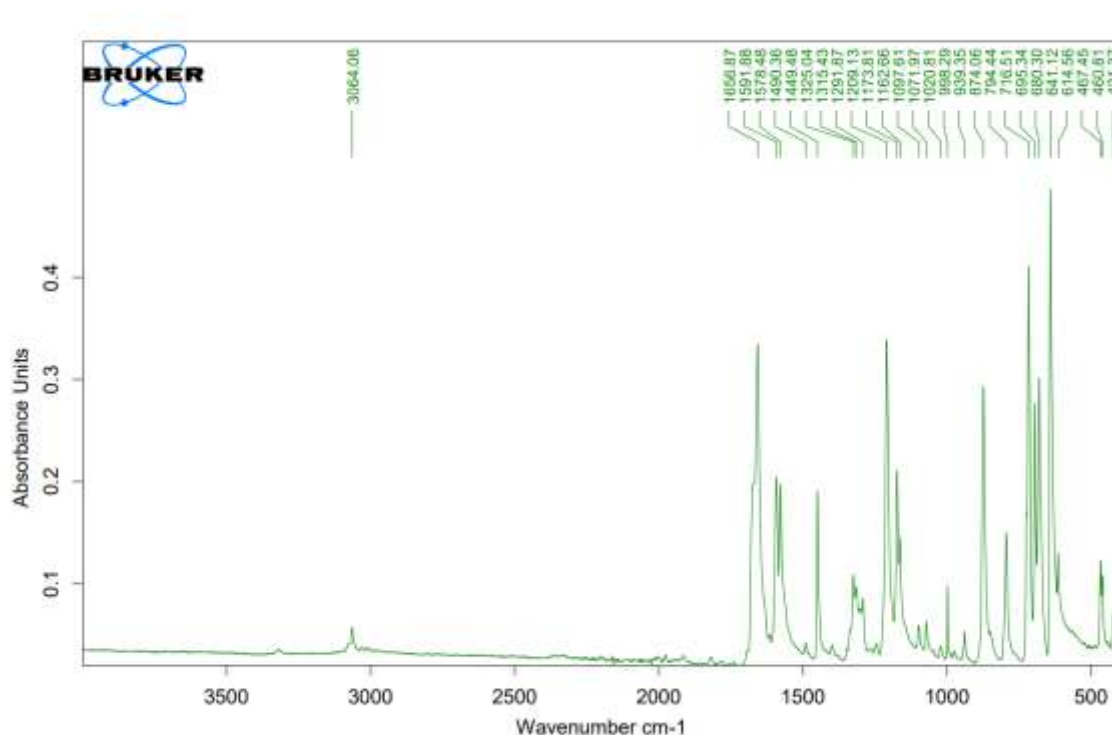
Wavenumber (cm ⁻¹)	Assignment
3376	O–H stretch
3059, 3028	Aromatic C–H stretch
2980, 2932	Aliphatic C–H stretch
1677	C=O stretch (conjugated ketone)
1594, 1577, 1490, 1448	Aromatic C=C stretch
1306, 1261, 1200, 1178	C–O stretch
753, 703	Monosubstituted aromatic C–H out-of-plane bend

Parameter	Value
1 Data File Name	c:/projects/data/2026/03/13/1156NMW/132930-1D PROTON-Unknown/Enhanced/data-1d
2 Spectrometer	Magitek Spinolve
3 Number of Scans	4
4 Acquisition Date	2026-03-13T13:29:36.951
5 Total acquisition time (min)	0.88
6 Nucleus	1H



δ (ppm)	Assignment
7.32–8.05	Aromatic H (10H)
5.97	Benzylic methine, PhCH(OH)C(O)Ph
4.14	Exchangeable O–H
1.23	Residual ethanol/solvent impurity

Experiment A2: Benzil



Wavenumber (cm ⁻¹)	Assignment
3064	Aromatic C–H stretch
1657	C=O stretch (α -diketone, conjugated)
1592, 1578, 1490, 1449	Aromatic C=C stretch
716, 695	Monosubstituted aromatic C–H out-of-plane bend

Although benzil contains two carbonyl groups, only a single C=O absorption was observed; this is a characteristic feature of symmetrical α -diketones, in which the symmetric in-phase stretch is IR-inactive due to the absence of a dipole moment change, and only the asymmetric stretch absorbs in the infrared region.

Aromatic C–H bending absorptions at 716 and 695 cm⁻¹ confirmed retention of the monosubstituted phenyl rings.

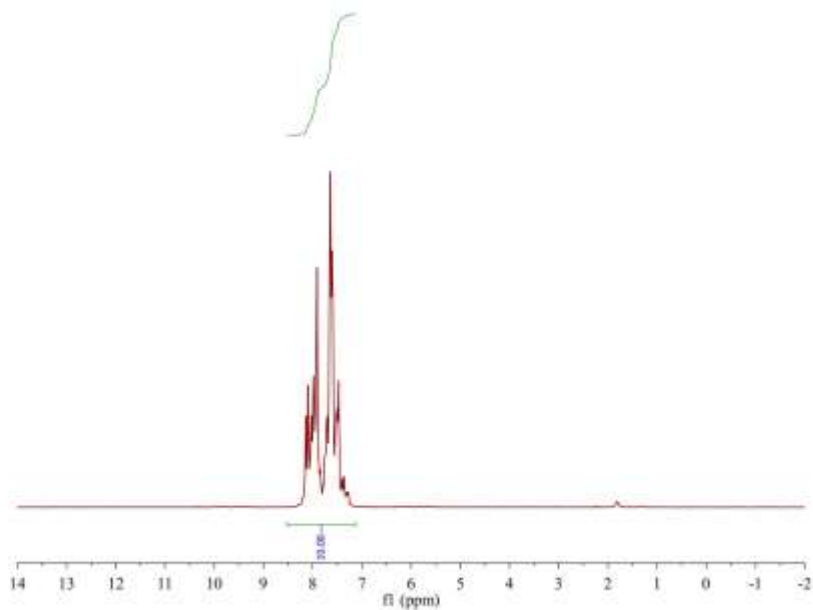


1D-1H-StandardScan

7.99 7.97 7.91 7.87 7.82 7.79 7.75 7.71 7.66 7.61 7.58 7.55 7.50 7.47 7.40 7.36 7.32 7.27

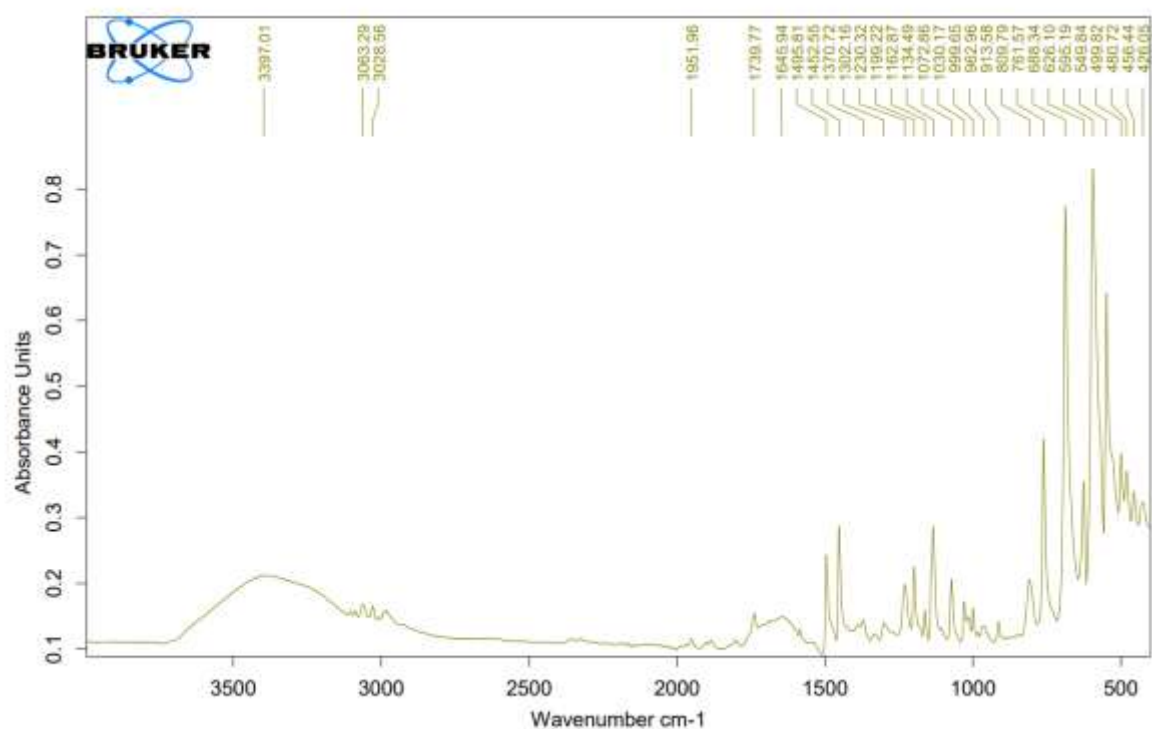
1.81

Parameter	Value
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2 Spectrometer:	Magritek Spinsolve
3 Number of Scans:	4
4 Acquisition Date:	2026-04-17T16:04:29.059
5 Total acquisition time (min):	0.00
6 Nucleus:	¹ H



δ (ppm)	Assignment
7.27–8.21	Aromatic H (10H, two phenyl rings)
1.81	Trace impurity/solvent

Experiment A3: Tetraphenylcyclopentadienone



Wavenumber (cm ⁻¹)	Assignment
3397	O–H stretch (residual moisture)
3063, 3029	Aromatic C–H stretch
1646	C=O stretch (cross-conjugated cyclopentadienone)
1496, 1453	Aromatic C=C stretch
762, 688	Monosubstituted aromatic C–H out-of-plane bend

Aromatic C–H bending absorptions at 761 and 688 cm⁻¹ confirmed retention of monosubstituted phenyl rings, while the broad weak absorption at 3397 cm⁻¹ was attributed to residual moisture rather than a structural hydroxyl group, given the absence of any corresponding O–H feature elsewhere in the spectrum or any aliphatic CH(OH) signal in the NMR.

1D PROTON

Sample : Unknown
Solvent : Chloroform
Custom : Unknown

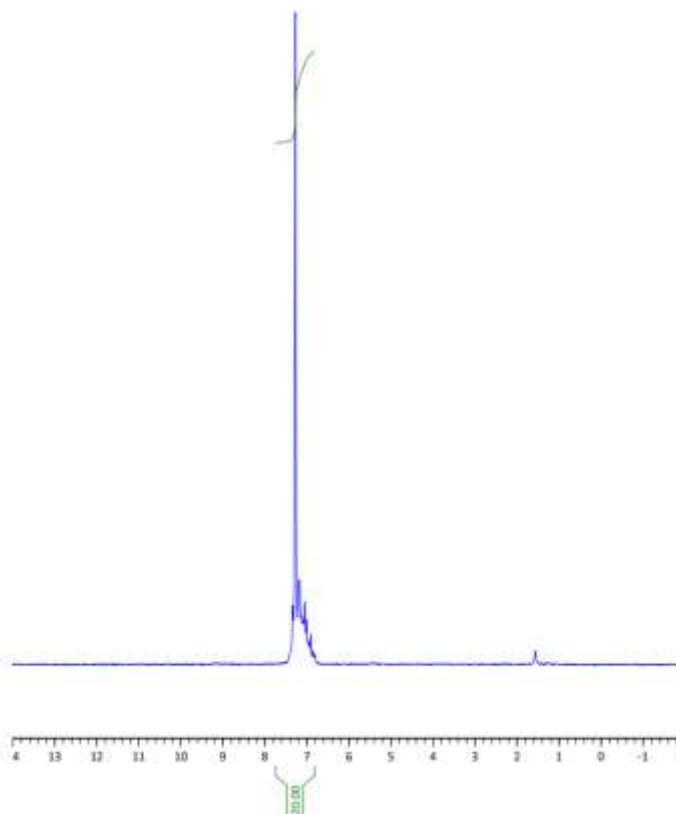
Acquisition Date : 2026-05-01 12:22:22
Number of scans : 4
Acquisition time : 6.5536 s
Repetition time : 15 s
Pulse angle : 90 degrees
Experiment Duration : 00:00:53

Processing
Resolution enhancement : 0.3 LB + 0.3 GB
Line broadening : 0.2 Hz
Phasing : P0 = 1.10 P1 = 0.00
Baseline correction : None

Meta data
Instrument : SPX1122
Instrument type : 43 CARBON
Software version : 2.0.5.4515
Acq. Windows user : L118NMR
Acq. Spinsolve user : Spinsolve
Data folder : c:\projects\data\2026\05\01\L118NMR\122221-1D PROTON-Unknown

Backup folder :
Last shim : 2026-05-01 10:44:43
Shim linewidth @ 50% : 0.42 Hz
Shim linewidth @ 0.55% : 10.73 Hz
Shim SNR : 39942

Integrals
ppm range Calibrated Absolute
7.75 - 6.82 20.00 266.57



δ (ppm)	Assignment
6.82–7.75	Aromatic H (20H, four phenyl rings)

Overall yield calculation:

How the overall yield calculation works

Imagine you started with 100 molecules of benzaldehyde:

Step A1 (47.91% yield):

- $100 \text{ benzaldehyde} \times 47.91\% = 47.91 \text{ benzoin molecules survive}$
- 52.09 molecules lost

Step A2 (70.92% yield):

- $47.91 \text{ benzoin} \times 70.92\% = 33.98 \text{ benzil molecules survive}$
- 13.93 more molecules lost

Step A3 (69.90% yield):

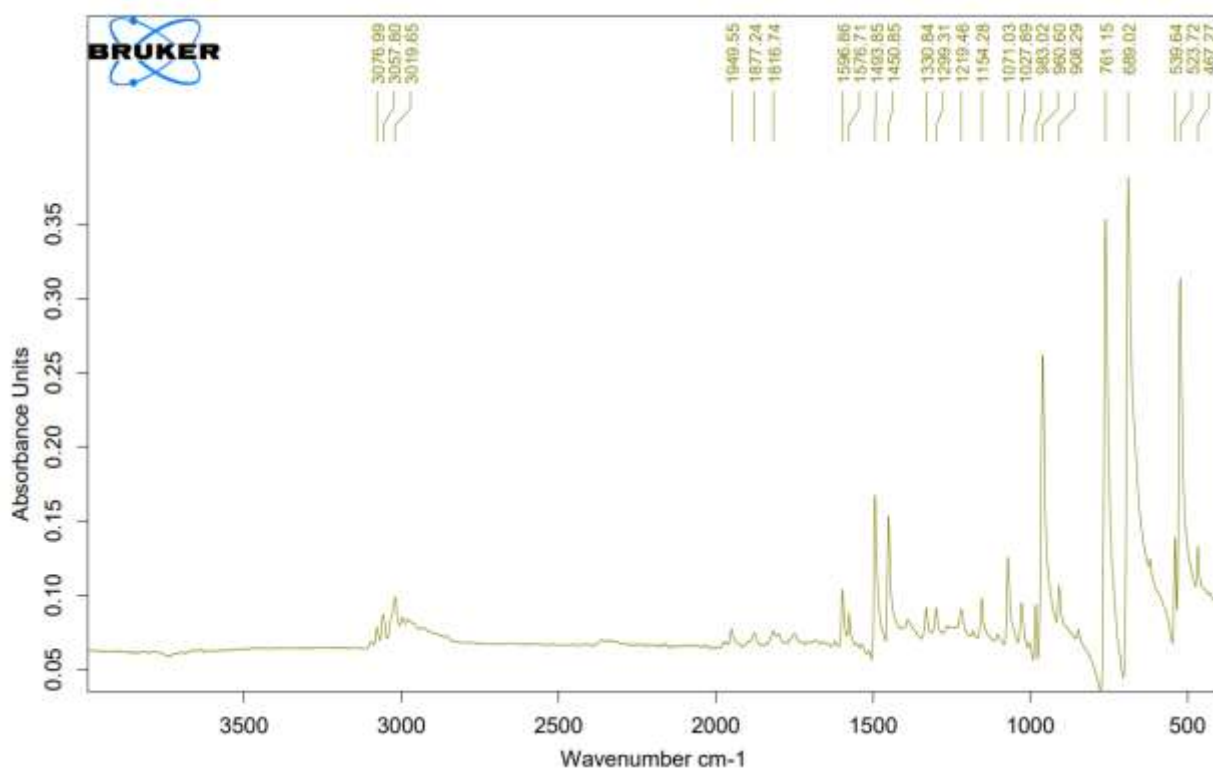
- $33.98 \text{ benzil} \times 69.90\% = 23.75 \text{ TPCPD molecules survive}$
- 10.23 more molecules lost

The math expressed as multiplication

$$\text{Overall yield} = 0.4791 \times 0.7092 \times 0.6990 = 0.2375 = 23.75\%$$

Multiply each step's yield (as a decimal) together because each step is *dependent* on the survival of material from the previous step.

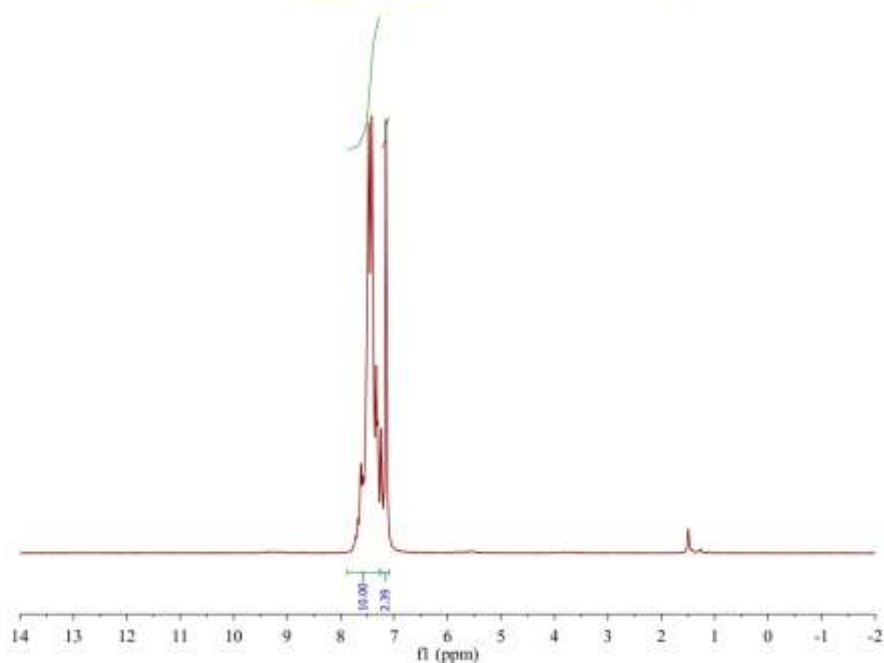
Experiment B1: (E)-Stilbene



Wavenumber (cm ⁻¹)	Assignment
3077, 3058, 3020	Aromatic + vinyl C–H stretch
1597, 1577, 1494	Aromatic C=C stretch
961	=C–H out-of-plane bend (<i>trans</i> -disubstituted alkene)
761, 689	Monosubstituted aromatic C–H out-of-plane bend

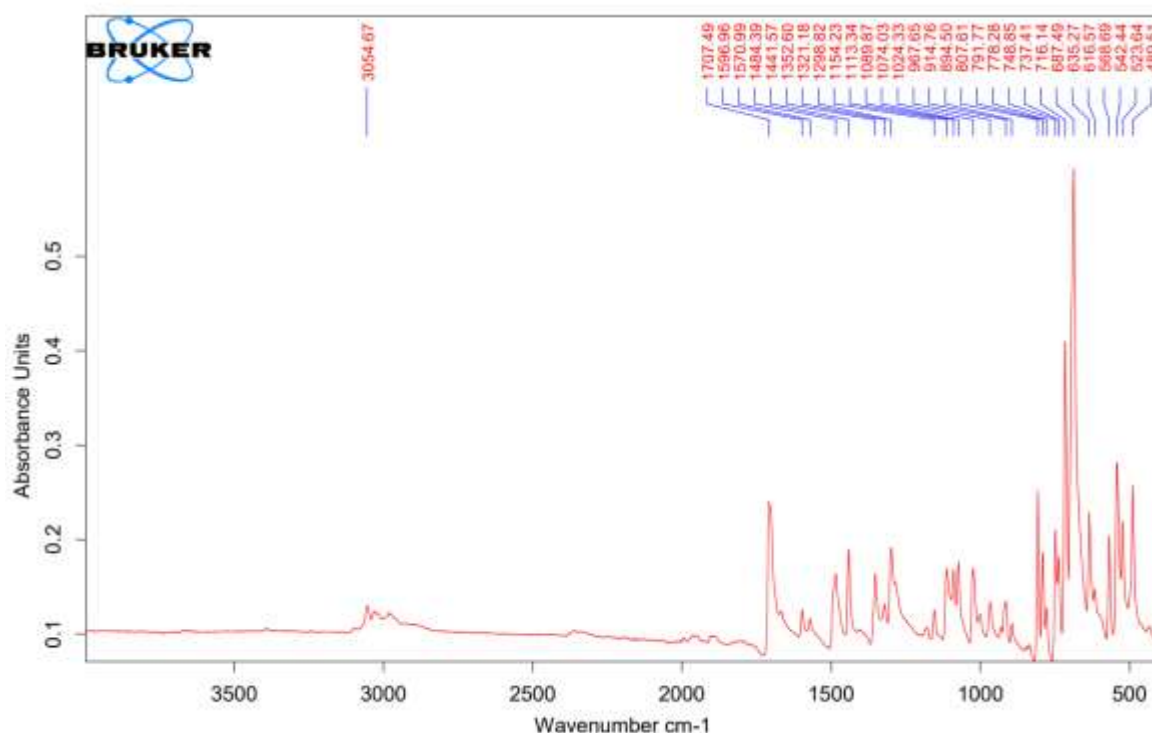
Aromatic C–H stretching absorptions at 3077, 3058, and 3020 cm⁻¹, the characteristic triplet of weak overtone bands at 1949, 1877, and 1817 cm⁻¹, and the strong out-of-plane bending modes at 761 and 689 cm⁻¹ confirmed that the phenyl rings in the product are monosubstituted, consistent with the structure of (E)-stilbene.

Parameter	Value
1 Data File Name	c:\projects\data\2026\03\27\1116h88\140924-1D-PROTON-Unknown\Enhanced\data_1d
2 Spectrometer	Magritek SpinSolve
3 Number of Scans	4
4 Acquisition Date	2026-03-27T14:09:25.093
5 Total acquisition time (min)	0.88
6 Nucleus	1H



δ (ppm)	Assignment
7.40–7.72	Aromatic H (10H, two phenyl rings)
7.15	Olefinic H (2H, <i>trans</i> C=C)
1.50	Residual water/trace impurity

Experiment B2: meso-Stilbene dibromide



Wavenumber (cm ⁻¹)	Assignment
3055	Aromatic C–H stretch
1597, 1571, 1484, 1442	Aromatic C=C stretch
1707	Trace glacial acetic acid (residual solvent)
748, 716, 687	Monosubstituted aromatic C–H out-of-plane bend
568, 542, 523	C–Br stretch

Aromatic C–H stretching at 3055 cm⁻¹, ring stretches at 1597, 1571, 1484 and 1442 cm⁻¹, and out-of-plane bending modes at 748, 716 and 687 cm⁻¹ collectively confirmed the integrity of the two monosubstituted phenyl substituents. A minor absorption at 1707 cm⁻¹ likely reflects trace glacial acetic acid retained from the reaction solvent, an impurity that could be addressed by more extensive vacuum drying of the product prior to characterisation; this impurity does not impact the structural conclusions.

1D PROTON

Sample : Unknown
Solvent : Chloroform
Custom : Unknown

Acquisition Date : 2026-05-01 12:17:18
Number of scans : 4
Acquisition time : 6.5536 s
Repetition time : 15 s
Pulse angle : 90 degrees
Experiment Duration : 00:00:53

Processing

Resolution enhancement : 0.3 LB + 0.3 GB
Line broadening : 0.2 Hz
Phasing : P0 = 3.50 P1 = 0.00
Baseline correction : None

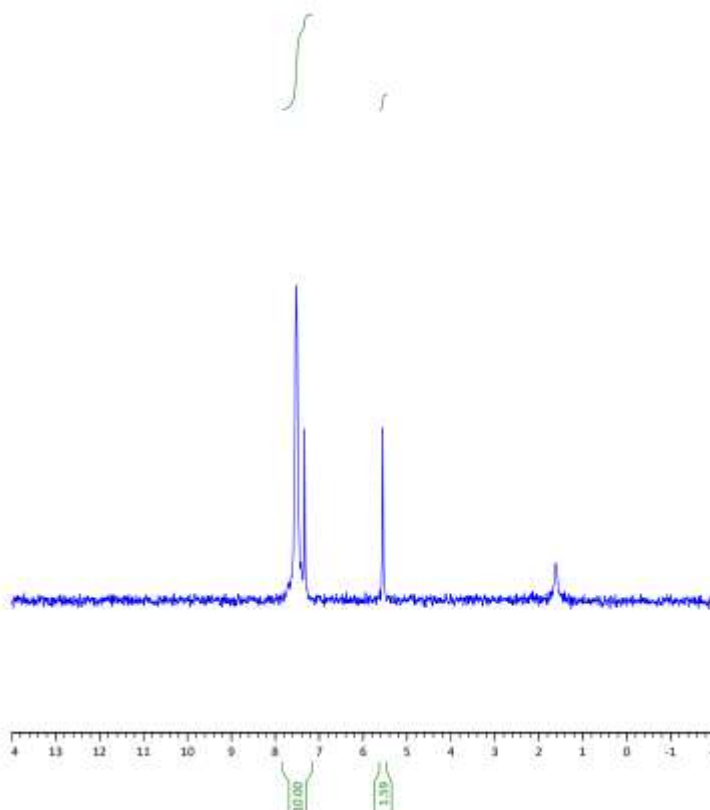
Meta data

Instrument : SPA1122
Instrument type : 41 CARBON
Software version : 2.0.5.4515
Acqu. Windows user : L116NMR
Acqu. Spinsolve user : Spinsolve
Data folder : c:\projects\data\2026\05\01\L116NMR\121717-1D PROTON-Unknown

Backup folder :
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Shim SNR : 39542

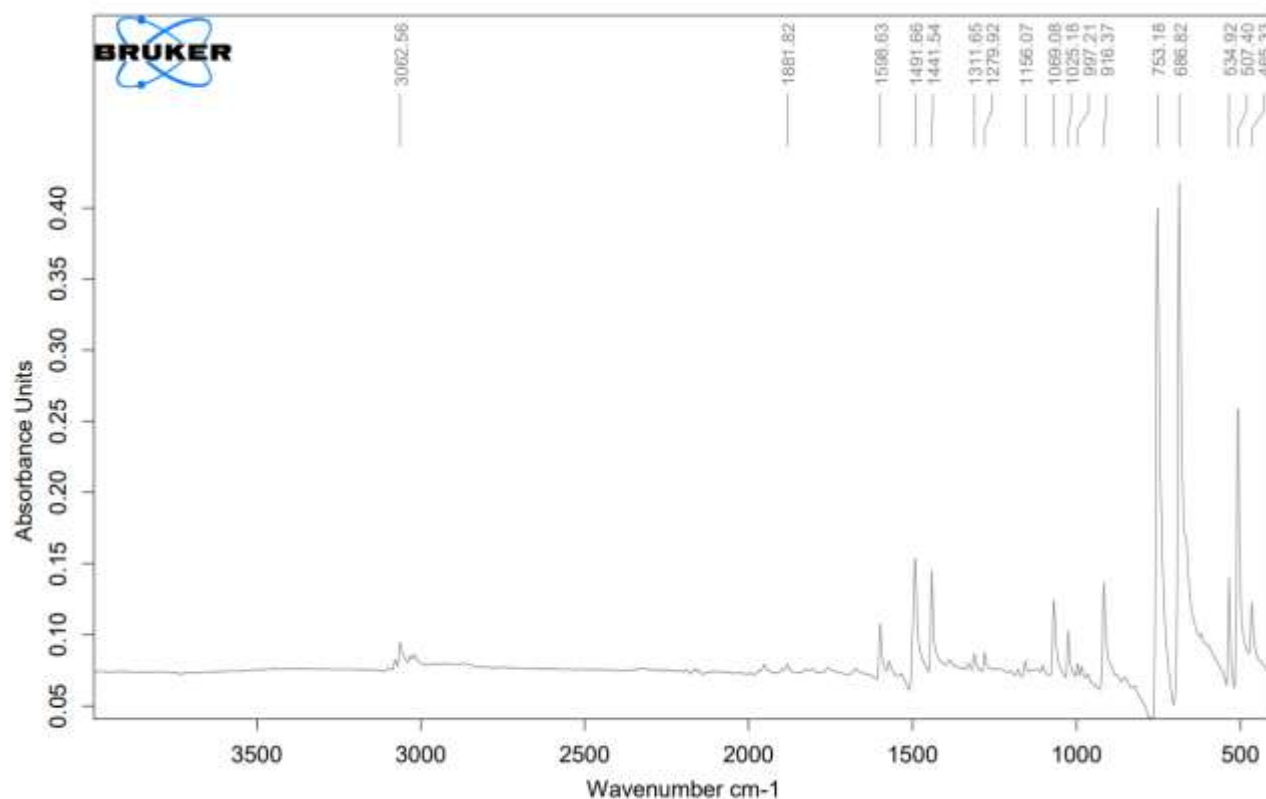
Integrals

ppm range	Calibrated	Absolute
5.63 - 5.46	1.59	6.83
7.83 - 7.16	10.00	42.87



δ (ppm)	Assignment
7.16–7.83	Aromatic H (10H, two phenyl rings)
5.46–5.63	CHBr methine (2H, equivalent by <i>meso</i> symmetry)

Experiment B3: Diphenylacetylene



Wavenumber (cm^{-1})	Assignment
3062	Aromatic C–H stretch
1598, 1491, 1441	Aromatic C=C stretch
753, 687	Monosubstituted aromatic C–H out-of-plane bend

Inversion centre:

In diphenylacetylene, the $\text{C}\equiv\text{C}$ triple bond stretch may not appear clearly in the IR spectrum because the molecule is highly symmetrical and has an inversion centre at the midpoint of the triple bond. An inversion centre is a point in a molecule where, for every atom on one side, an identical atom exists at the same distance in the exact opposite direction. In diphenylacetylene, the left side of the molecule mirrors the right side, making it centrosymmetric. IR spectroscopy only detects vibrations that cause a change in the molecule's net dipole moment. During the symmetric $\text{C}\equiv\text{C}$ stretching vibration, any dipole change on one side is cancelled by an equal and opposite change on the other side, so the net dipole change is zero. As a result, the $\text{C}\equiv\text{C}$ stretch is IR-inactive or very weak, even though the triple bond is present.

Mutual exclusion principle. This is a general rule for centrosymmetric molecules: any vibration that is IR-active is Raman-inactive, and any vibration that is Raman-active is IR-inactive. The two techniques are "mutually exclusive." The C≡C stretch in diphenylacetylene falls in the second category — it's invisible to IR but would be a strong, sharp band in a Raman spectrum.

1D PROTON

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Solvent : Chloroform
Custom : Unknown

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Repetition time : 15 s
Pulse angle : 90 degrees
Experiment Duration : 00:00:53

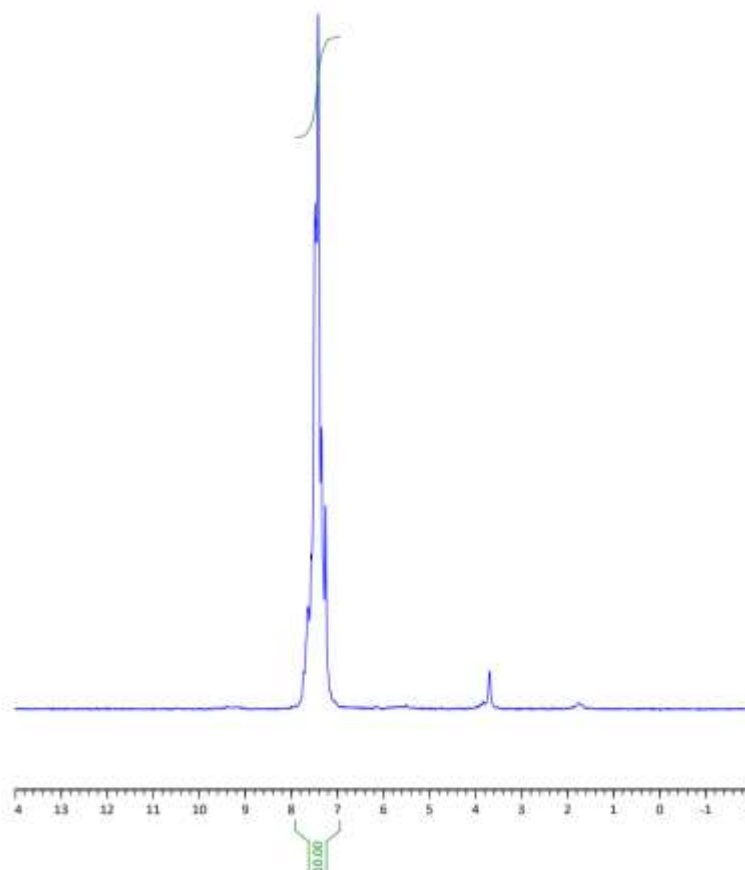
Processing
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Line broadening : 0.2 Hz
Phasing : PQ = 1.50 P1 = 0.00
Baseline correction : None

Meta data
Instrument : SPA1122
Instrument type : 43 CARBON
Software version : 2.0.5.4515
Acqu. Windows user : L116NMR
Acqu. Spinsolve user : Spinsolve
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Backup folder :
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Shim SNR : 38077

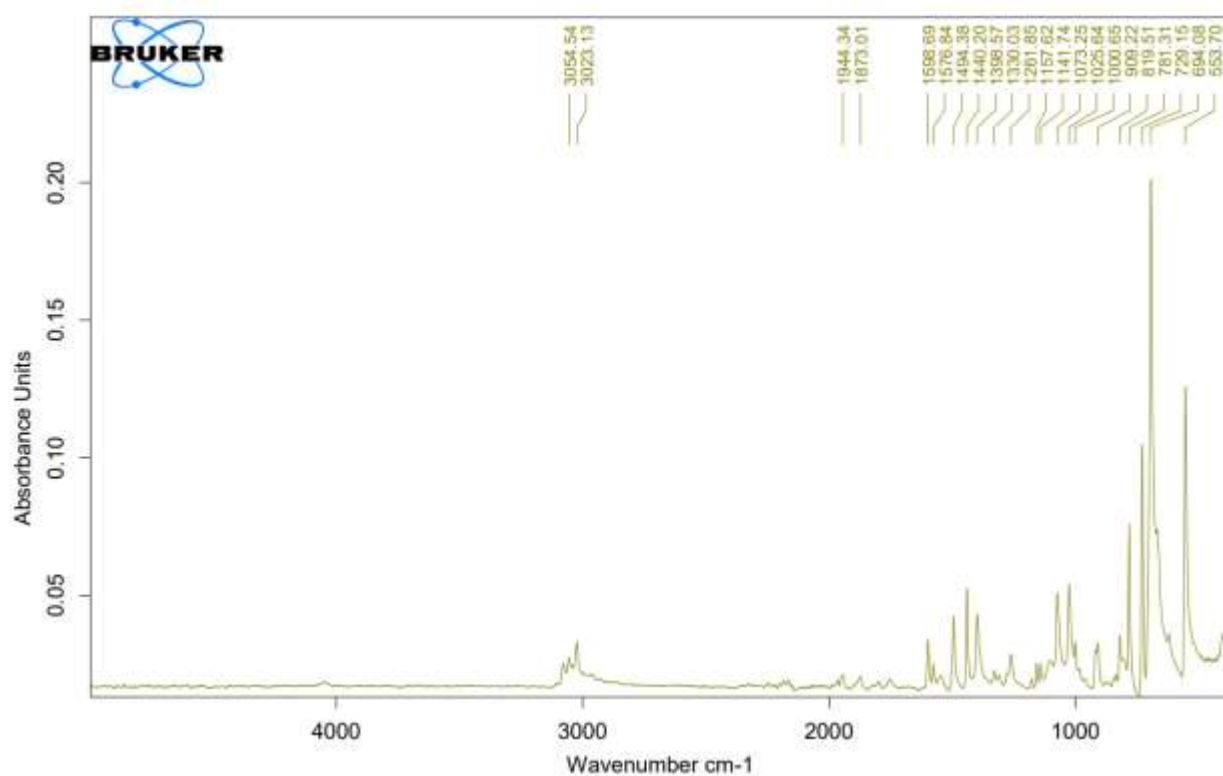
Integrals

ppm range	Calibrated	Absolute
7.92 - 6.95	10.00	852.01



δ (ppm)	Assignment
6.95–7.92	Aromatic H (10H, two equivalent phenyl rings)

Experiment AB4: Hexaphenylbenzene



Wavenumber (cm ⁻¹)	Assignment
3054, 3023	Aromatic C–H stretch
1598, 1576, 1494, 1440	Aromatic C=C stretch
729, 694	Monosubstituted aromatic C–H out-of-plane bend

1D PROTON

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Solvent : Chloroform
Custom : Unknown

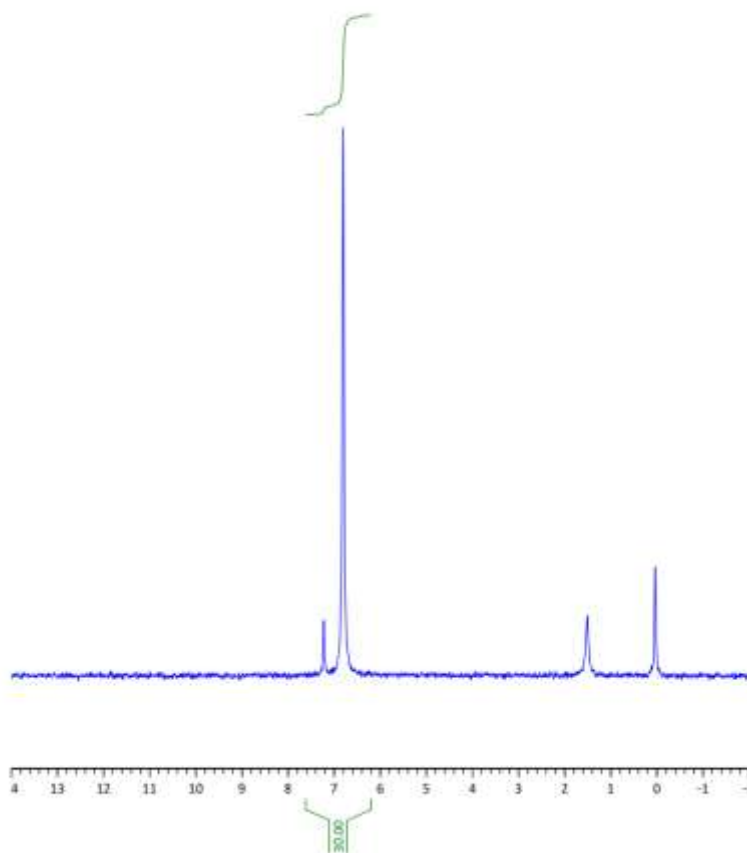
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Number of scans : 4
Acquisition time : 6.5536 s
Repetition time : 15 s
Pulse angle : 90 degrees
Experiment Duration : 00:00:53

Processing
Resolution enhancement : 0.3 LB + 0.3 GB
Line broadening : 0.2 Hz
Phasing : PQ = 0.90 P1 = 0.00
Baseline correction : None

Meta data
Instrument : SPA1122
Instrument type : 43 CARBON
Software version : 2.0.5.4515
Acqu. Windows user : LI16NMR
Acqu. Spinsolve user : Spinsolve
Data folder : c:\projects\data\2026\05\15\L1
16NMR\130142-1D PROTON-Unknown

Backup folder :
Last shim : 2026-05-13 10:08:46
Shim linewidth @ 50% : 0.39 Hz
Shim linewidth @ 0.55% : 11.04 Hz
Shim SNR : 38077

Integrals
ppm range Calibrated Absolute
7.62 - 6.20 30.00 82.79



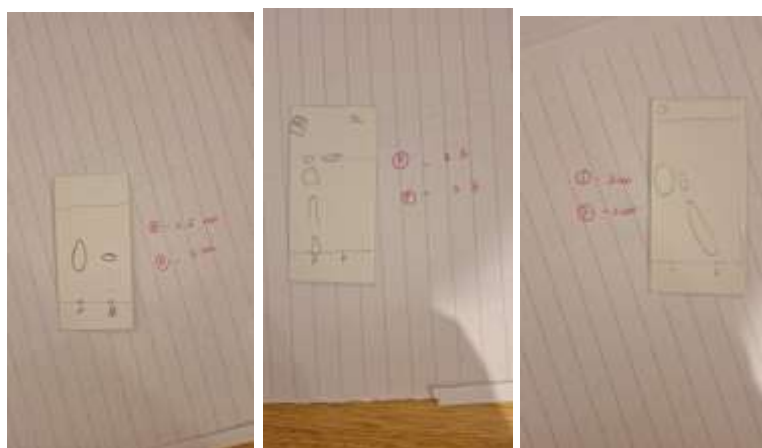
δ (ppm)	Assignment
6.20–7.62	Aromatic H (30H, six equivalent outer phenyl rings)

The workup:

In the AB4 reaction, diphenylacetylene was used in 2.2-fold excess, meaning that a significant amount of unreacted diphenylacetylene was expected to remain in the cooled reaction mixture alongside the hexaphenylbenzene product and silicone oil. The workup therefore relies on differences in solubility to separate the two compounds. Diphenylacetylene is a smaller, relatively non-polar and low-melting compound that dissolves readily in non-polar organic solvents such as petroleum spirits. In contrast, hexaphenylbenzene is much less soluble in petroleum spirits at room temperature, despite also being a non-polar aromatic hydrocarbon. This low solubility is mainly due to its much higher molecular weight, bulky rigid structure and strong crystal-lattice interactions, which also contribute to its extremely high melting point. As a result, washing the cooled reaction mixture with petroleum spirits dissolves and removes excess diphenylacetylene, while hexaphenylbenzene remains as a solid and can be collected by filtration.

If TPCPD had been used in excess, the final product would likely have been contaminated with unreacted purple TPCPD, which would be difficult to remove because it is much less soluble than diphenylacetylene in petroleum spirits and has a relatively similar size and structure to HPB. This could require further purification by chromatography or recrystallisation. Therefore, diphenylacetylene was used in excess instead, because any unreacted diphenylacetylene is easier to remove by washing with petroleum spirits, while the more difficult-to-remove TPCPD is consumed during the reaction.

TLC:



A1

B1

A2

Conclusion

The six outer phenyl rings in hexaphenylbenzene cannot rotate independently because the hydrogen atoms at their ortho positions physically clash with the ortho hydrogens of neighbouring rings. As a result, rotation of one phenyl ring requires simultaneous movement of the adjacent rings, similar to meshed gears in which turning one cog forces the neighbouring cog to rotate in the opposite direction.

Chemical structures and reaction schemes were drawn using ChemDraw.